

The development of cities and regions

Balázs Lengyel

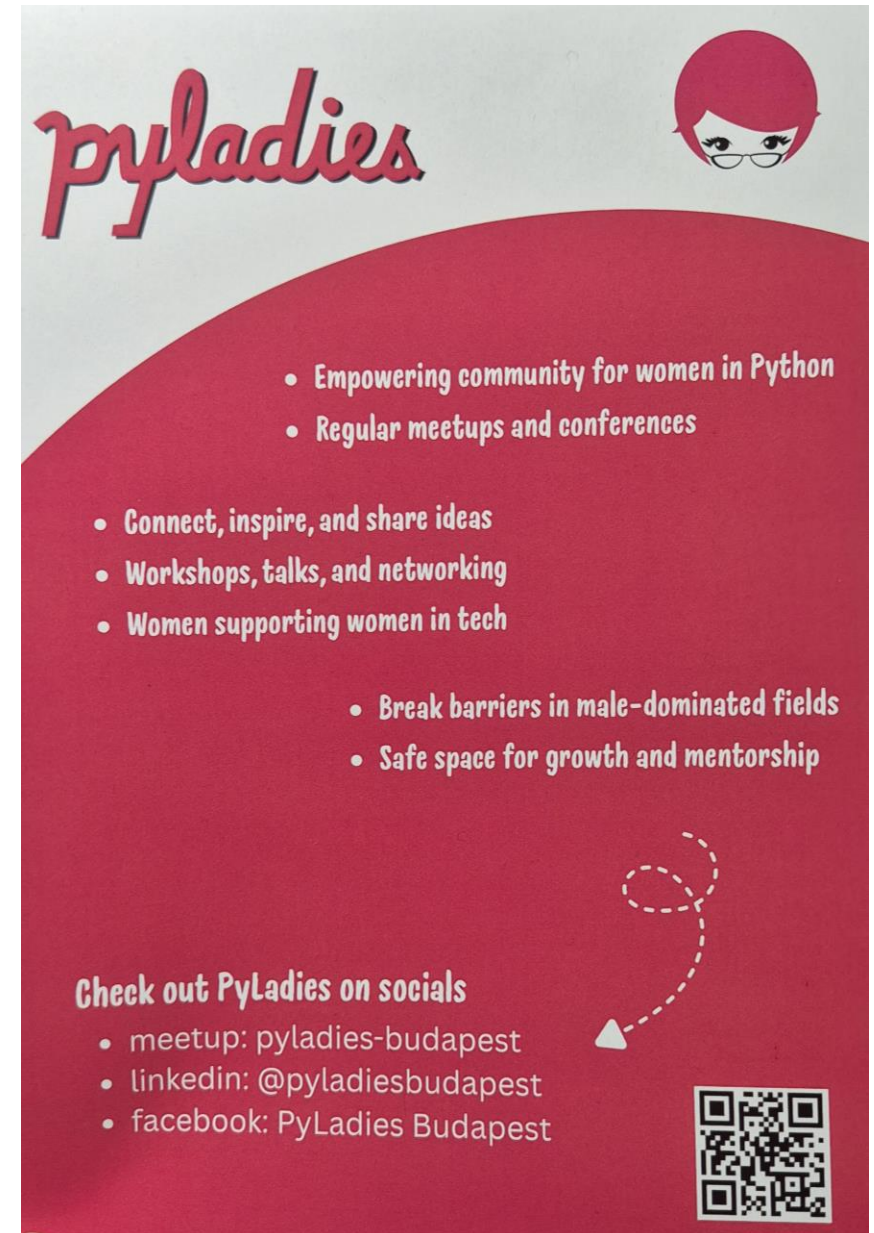
Economic Complexity, Class 4 – Oct 14 2025

Social Data Science, Corvinus University of Budapest

Test: Audretsch and Feldman, AER 1996

The community matters

- ANETI seminars every Thursday at 3pm
anetilab.org
- Department of Network Science
Social Lectures Night in November
- TEAMS!!! Cloud_SDS_25-26 !



The poster for PyLadies Budapest features a light grey background at the top with the word "pyladies" in a red, cursive font. To the right is a stylized illustration of a woman with short red hair and glasses. The bottom half of the poster is a large red curved shape. Inside this shape, there are two columns of bullet points in white text. At the bottom left, there is a section titled "Check out PyLadies on socials" followed by three bullet points for meetup, linkedin, and facebook. At the bottom right, there is a QR code and a dashed white line with an arrow pointing towards it.

pyladies

- Empowering community for women in Python
- Regular meetups and conferences
- Connect, inspire, and share ideas
- Workshops, talks, and networking
- Women supporting women in tech
- Break barriers in male-dominated fields
- Safe space for growth and mentorship

Check out PyLadies on socials

- meetup: pyladies-budapest
- linkedin: @pyladiesbudapest
- facebook: PyLadies Budapest



Student helpers are needed: geoinno2026.org

GEQINNO2026

[ABOUT ▾](#)[PARTICIPATE ▾](#)[VISITOR INFO ▾](#)[CONTACT](#)

28th - 30th January 2026
Corvinus University of Budapest



Beata Javorcik

European Bank for Reconstruction and Development



Lars Coenen

Western Norway University of Applied Sciences & University of Bergen



Luisa Gagliardi

Bocconi University



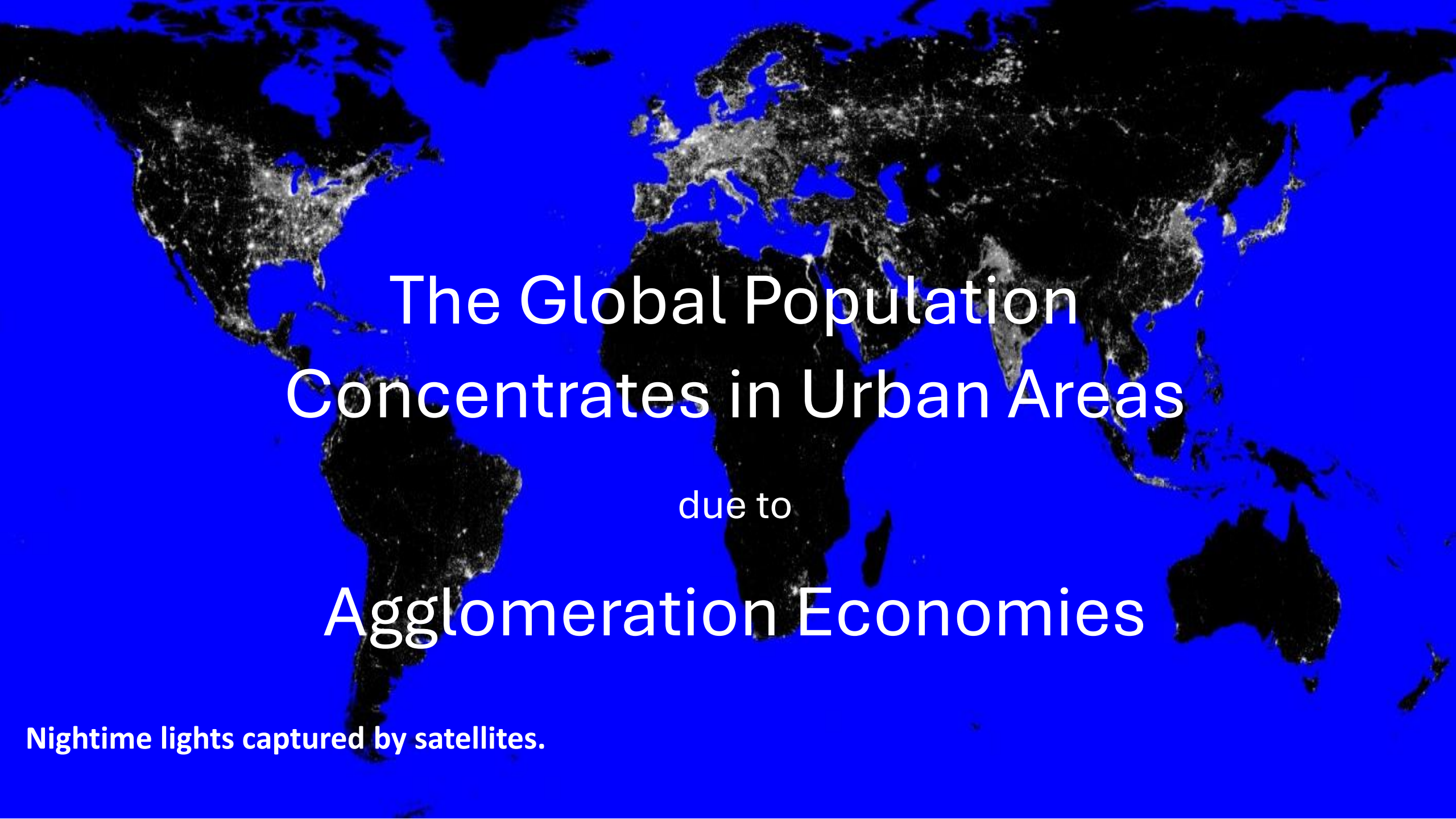
Lee Fleming

University of California, Berkeley

Jobs at the ANETI Lab

2 research assistant positions

- 10 hours a week (potential increase in the summer)
- 10 months: 2025 November – 2026 August
- Project 1: „Work-from-home in software development”. Data: Slack, Github, Linear records of a mid-size company.
- Project 2: „Social media communication in large cities”. Data: geo-located Twitter in London, Amsterdam, US cities.

A world map with a black background, where landmasses are outlined in white. The map is filled with numerous small, bright white dots and clusters of dots, representing nighttime lights captured by satellites. These lights are most concentrated in North America, Europe, and East Asia, with smaller clusters in South America, Africa, and Australia. The text is overlaid on the map.

The Global Population Concentrates in Urban Areas

due to

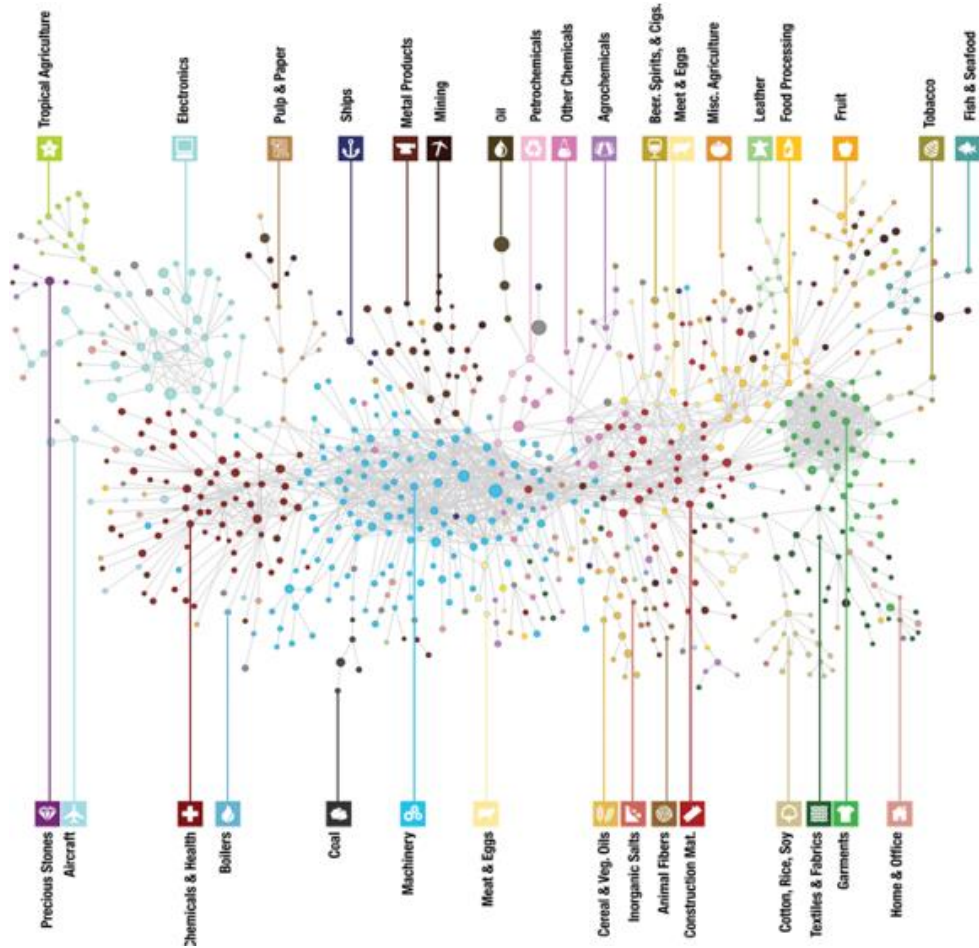
Agglomeration Economies

Nighttime lights captured by satellites.

1992



Product space: assumptions and messages



- Capabilities/knowledge
- Spatial specialization

Previous class



Explicit Knowledge

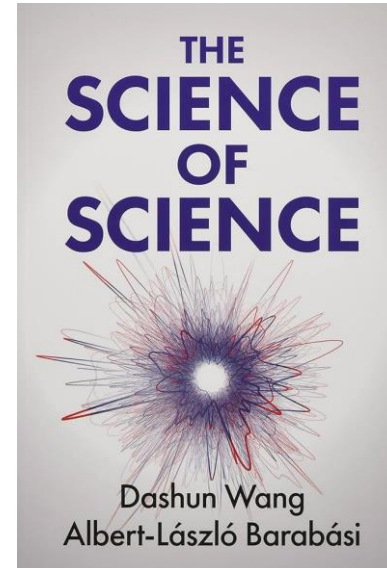
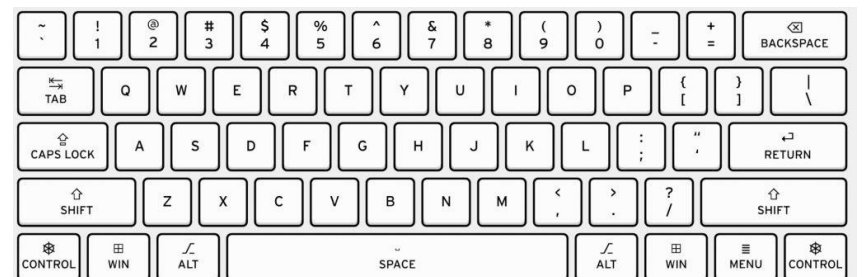
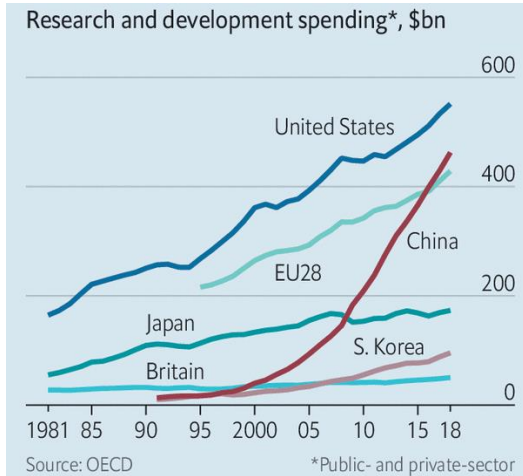
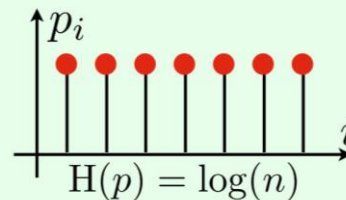
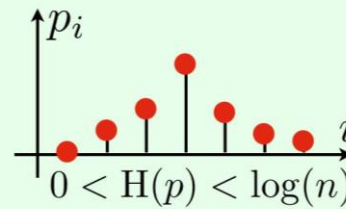
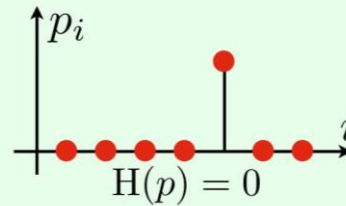
- Declarative information
- Rules & procedures

Tacit Knowledge

- Pattern recognition
- Perceptual discrimination
- Judging typicality
- Mental models
- Mindsets



$$H(p) \stackrel{\text{def.}}{=} - \sum_i p_i \log(p_i)$$



The second block

3	Oct 7	Balázs Lengyel Knowledge, innovation, and technological development	II. Innovation and Economic Geography	[4]
4	Oct 14	Balázs Lengyel The development of cities and regions: agglomeration economies, the specialization-diversity debate		[5,6]
5	Oct 21	Zoltán Elekes Labor markets and skills: labor mobility, skill-relatedness		[7,8]
6	Nov 4	Balázs Lengyel Learning and innovation in networks: knowledge sourcing and dynamics of networks		[9,10]
7	Nov 11	Balázs Lengyel and Sándor Juhász Spatial and network inequalities: social capital, segregation and mixing		[11,12]
8	Nov 18	Presentation of students' research proposal		

[4] Dosi, G. (1988). Sources, procedures, and microeconomic effects of innovation. *Journal of Economic Literature*, 1120-1171.

[5] Audretsch, D. B., & Feldman, M. P. (1996). R&D spillovers and the geography of innovation and production. *The American Economic Review*, 86(3), 630-640.

[6] Glaeser, E. L., Kallal, H. D., Scheinkman, J. A., & Shleifer, A. (1992). Growth in cities. *Journal of Political Economy*, 100(6), 1126-1152.

[7] Neffke, F., Henning, M., & Boschma, R. (2011). How do regions diversify over time? Industry relatedness and the development of new growth paths in regions. *Economic Geography*, 87(3), 237-265.

[8] Alabdulkareem, A., Frank, M. R., Sun, L., AlShebli, B., Hidalgo, C., & Rahwan, I. (2018). Unpacking the polarization of workplace skills. *Science Advances*, 4(7), eaao6030.

[9] Sorenson, O., Rivkin, J. W., & Fleming, L. (2006). Complexity, networks and knowledge flow. *Research Policy*, 35(7), 994-1017.

[10] Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, 52(10), 104886.

[11] Tóth, G., Wachs, J., Di Clemente, R., Jakobi, Á., Ságvári, B., Kertész, J., & Lengyel, B. (2021). Inequality is rising where social network segregation interacts with urban topology. *Nature Communications*, 12(1), 1143.

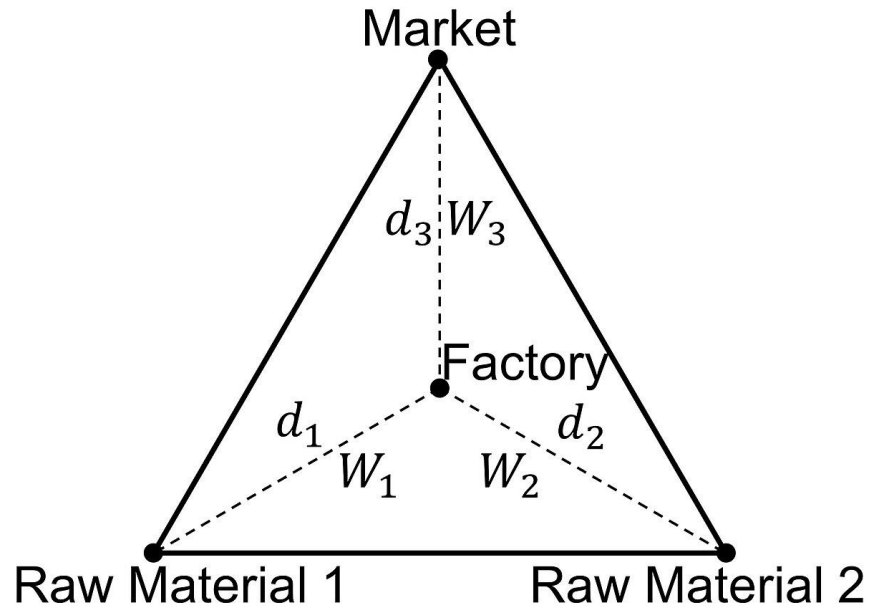
[12] Aiello, L. M., Vybornova, A., Juhász, S., Szell, M., & Bokányi, E. (2025). Urban highways are barriers to social ties. *PNAS*, 122(10), e2408937122.

Today

1. Location theory: costs and markets
2. Centrum and hierarchy
3. Agglomeration economies
4. Concentration of knowledge
5. Industry structure: specialization versus diversity

1. Location theory

Alfred Weber's location triangle



Looking for the minimal transportation cost location

$$TC = W_1 d_1 + W_2 d_2 + W_3 d_3$$

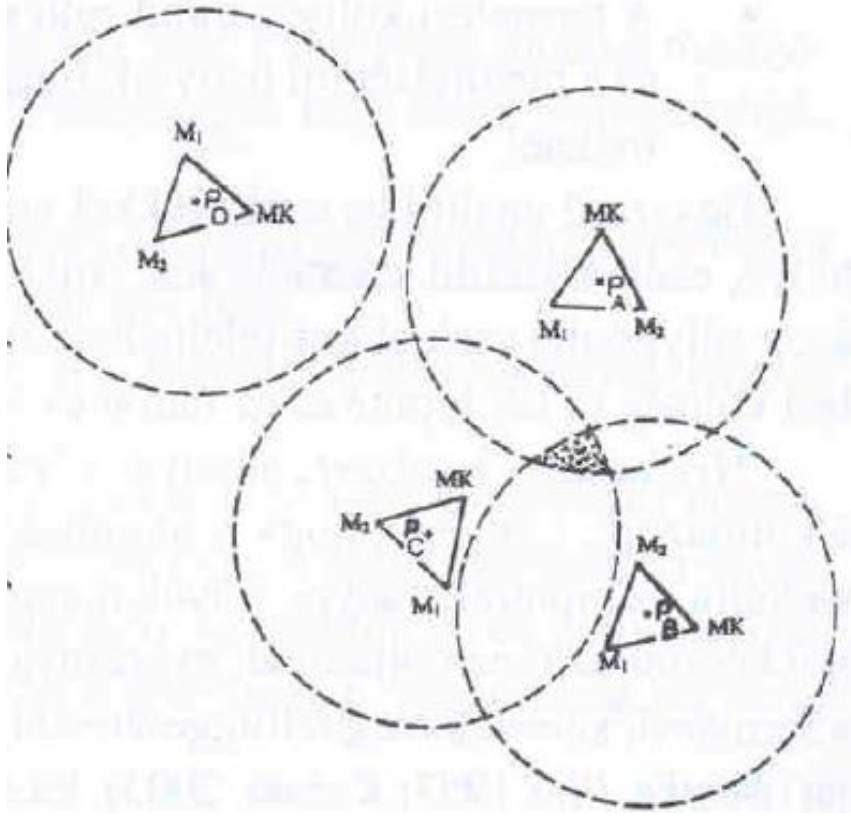
(W: weights)
(d: distance)

Ha

$$W_3 d_3 > W_1 d_1 + W_2 d_2 \Rightarrow \text{towards market}$$

$$W_3 d_3 < W_1 d_1 + W_2 d_2 \Rightarrow \text{towards raw materials}$$

Weber: Relocation to agglomeration

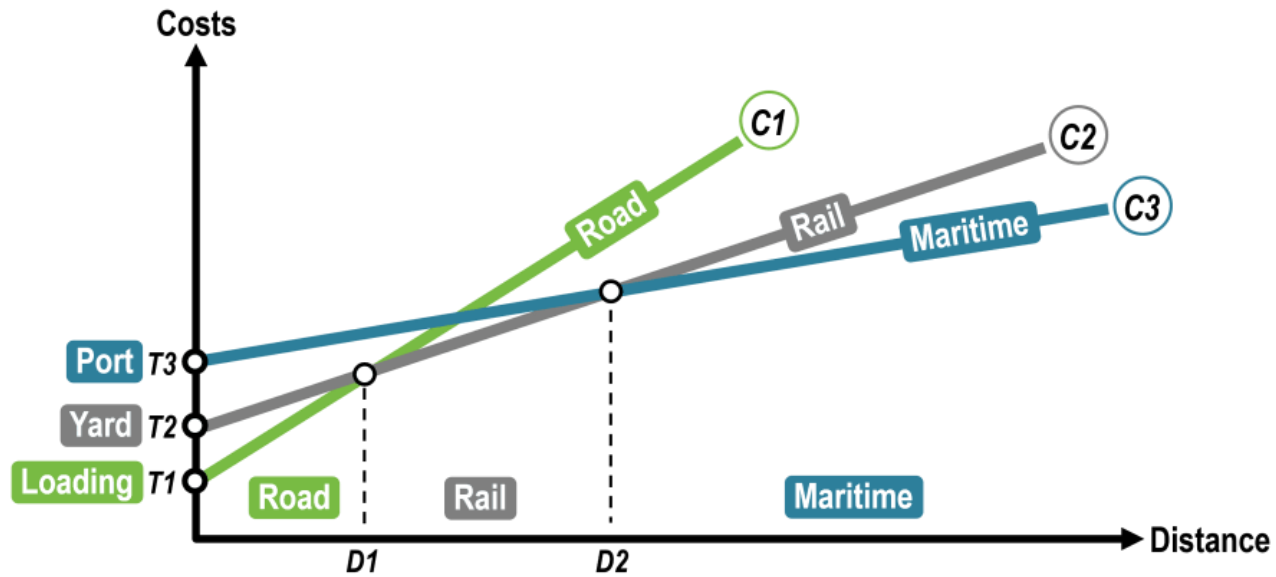


Circles (isodapan): constant additional costs

Shaded area: agglomeration where labor is available

A firm locates to an agglomeration area if agglomeration benefits are higher than additional transport costs.

Lösch: cost space and markets

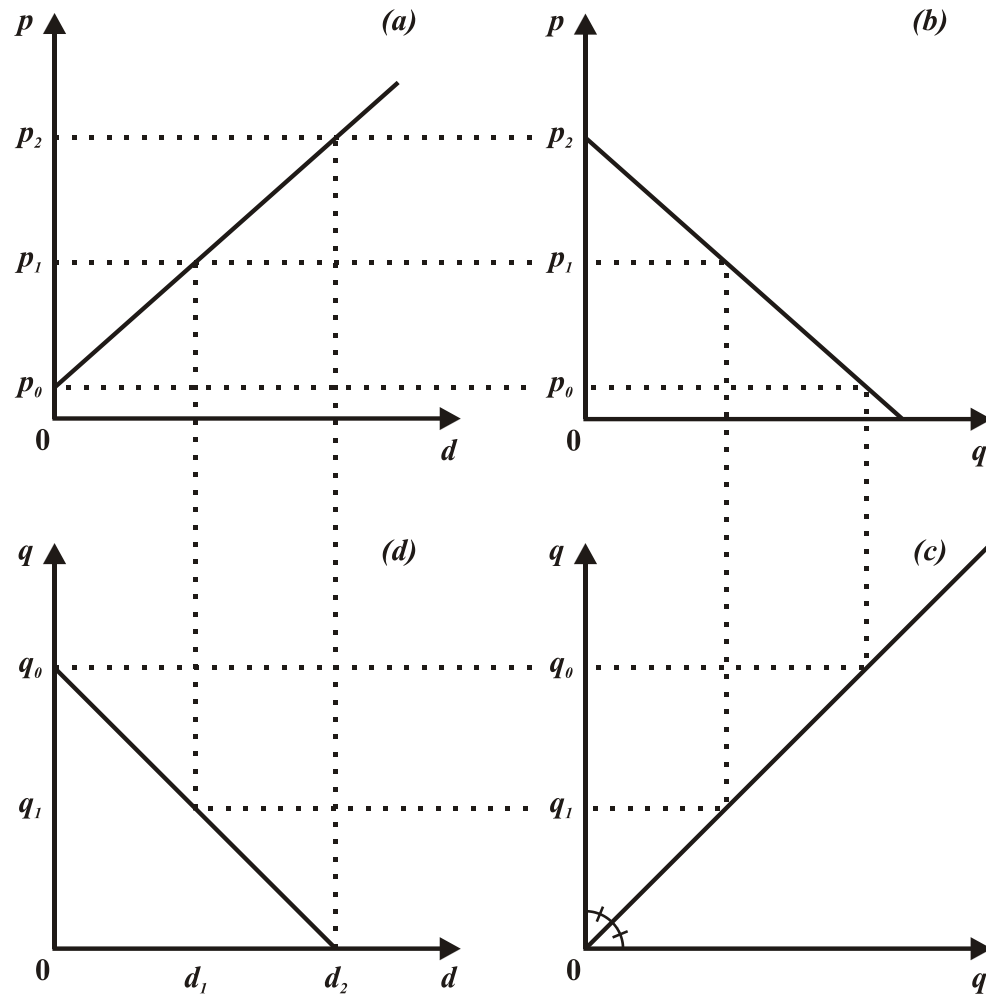


Final cost is the sum of

1. Free on Board (FOB):
production costs + loading
costs

2. Transportation costs to
markets

Lösch: spatial demand



(a) Supply curve of the firm

$$p = p^* + \tau d$$

(b) Demand curve of the customer

$$p = a - b x_i$$

(c) lead-through diagram

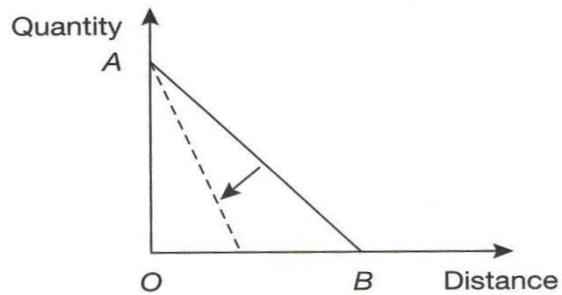
(d) Quantity – distance curve

$$x_i = (a - p^*)/b - \tau d/b$$

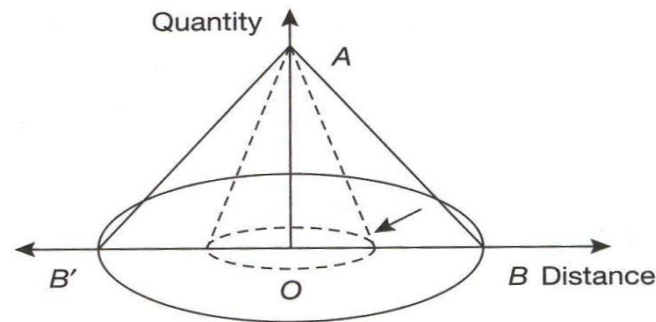
Lösch: spatial markets

(a) Increase in transportation costs

(a1) linear market

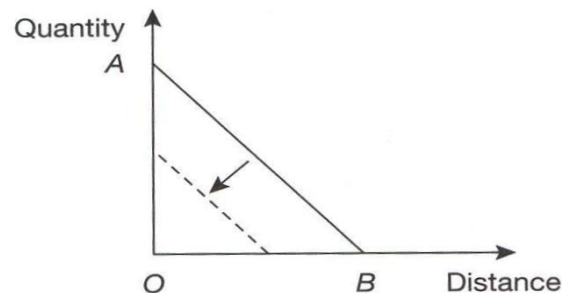


(a2) circular market (volume of the Lösch cone)

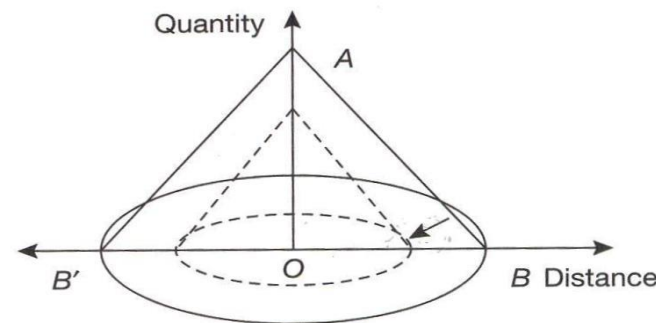


(b) Increase in the price of the good

(b1) linear market

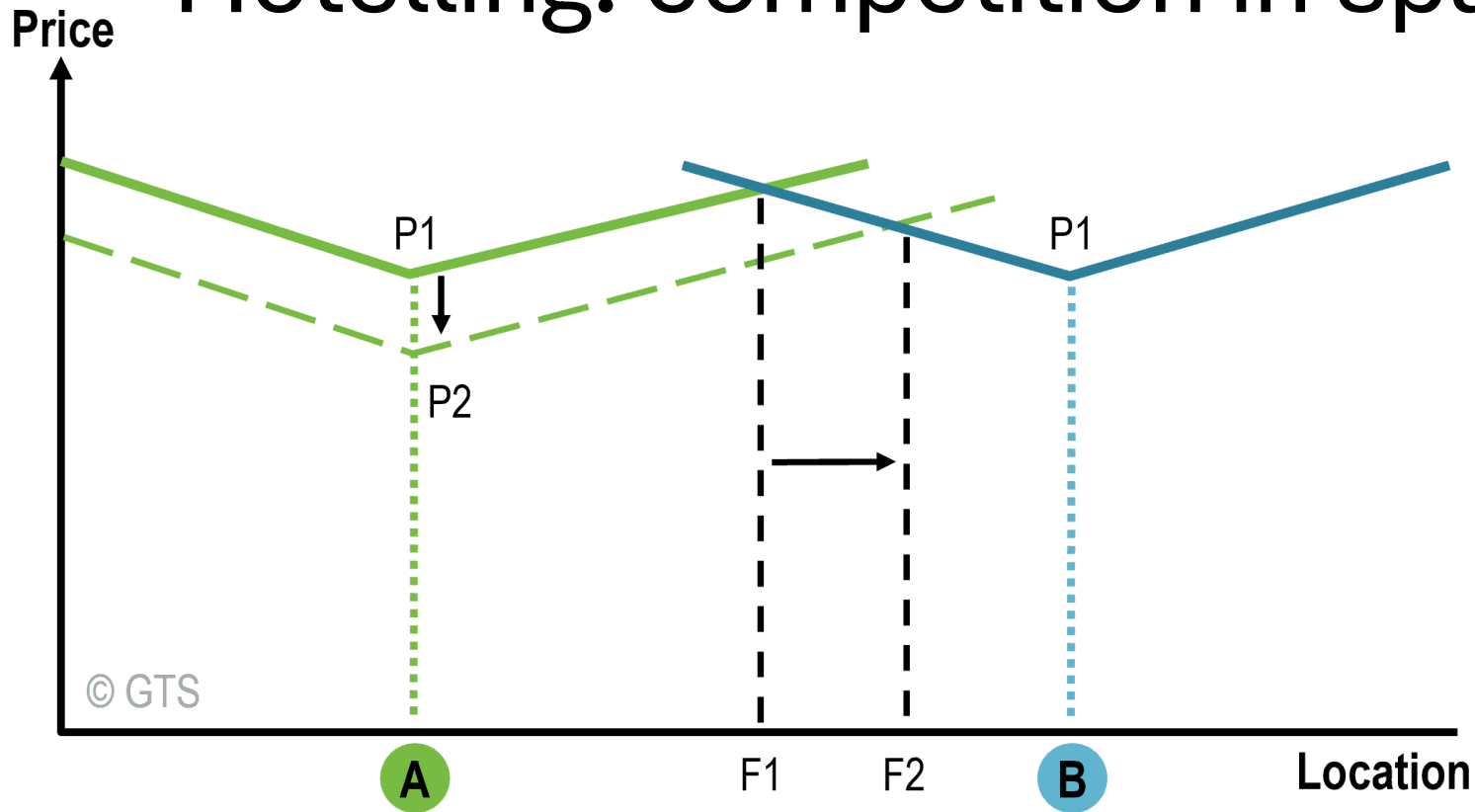


(b2) circular market (volume of the Lösch cone)

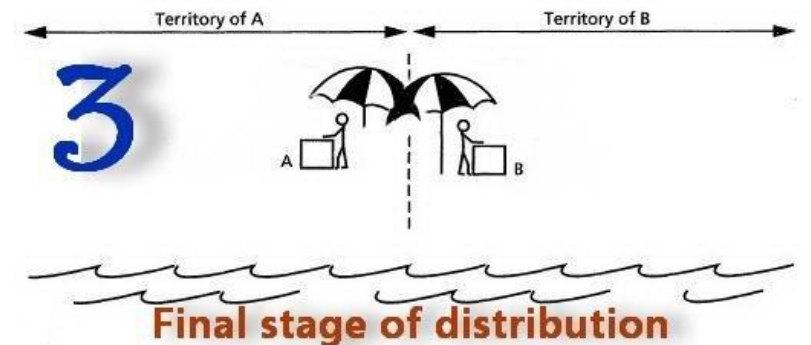
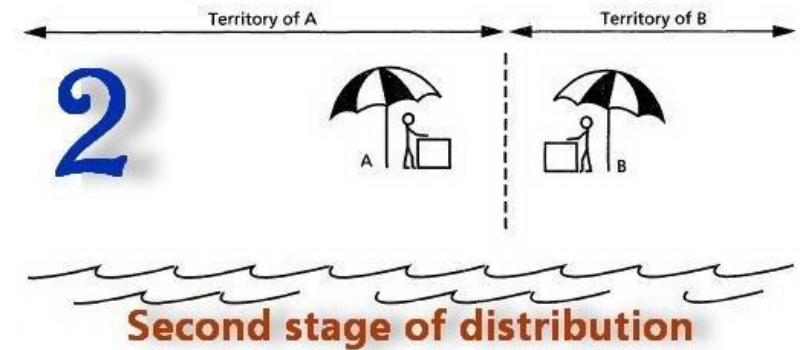
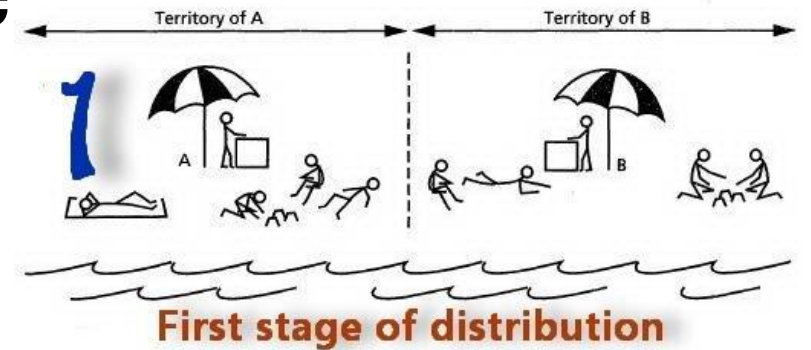


Increasing transportation costs and increasing product price leads to shrinking market size

Hotelling: competition in space

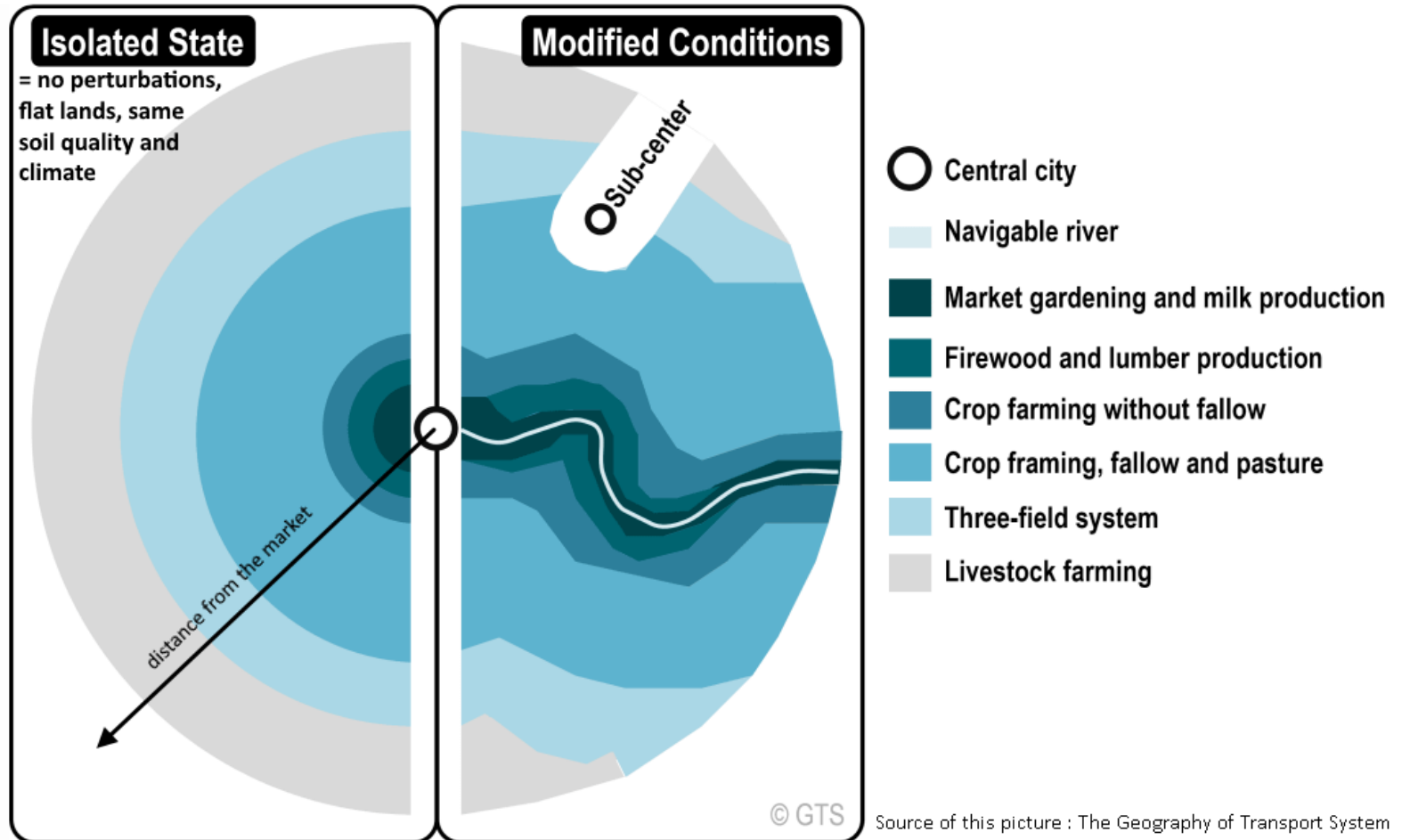
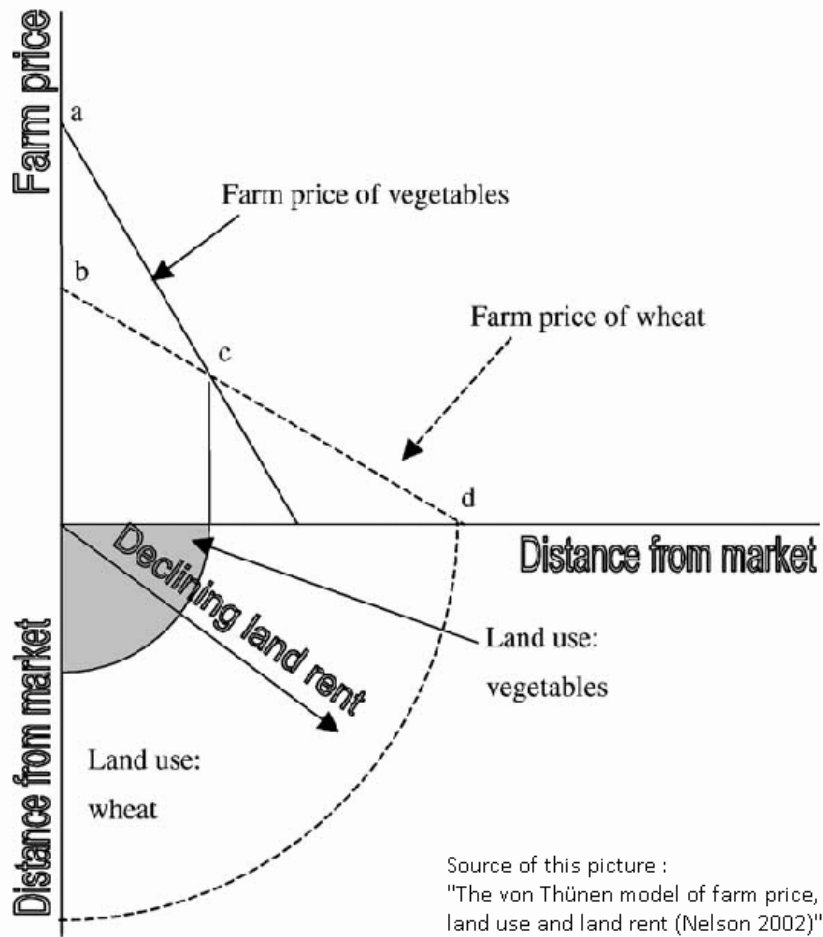


Firms co-locate to maximize market share



2. Centrum and hierarchy

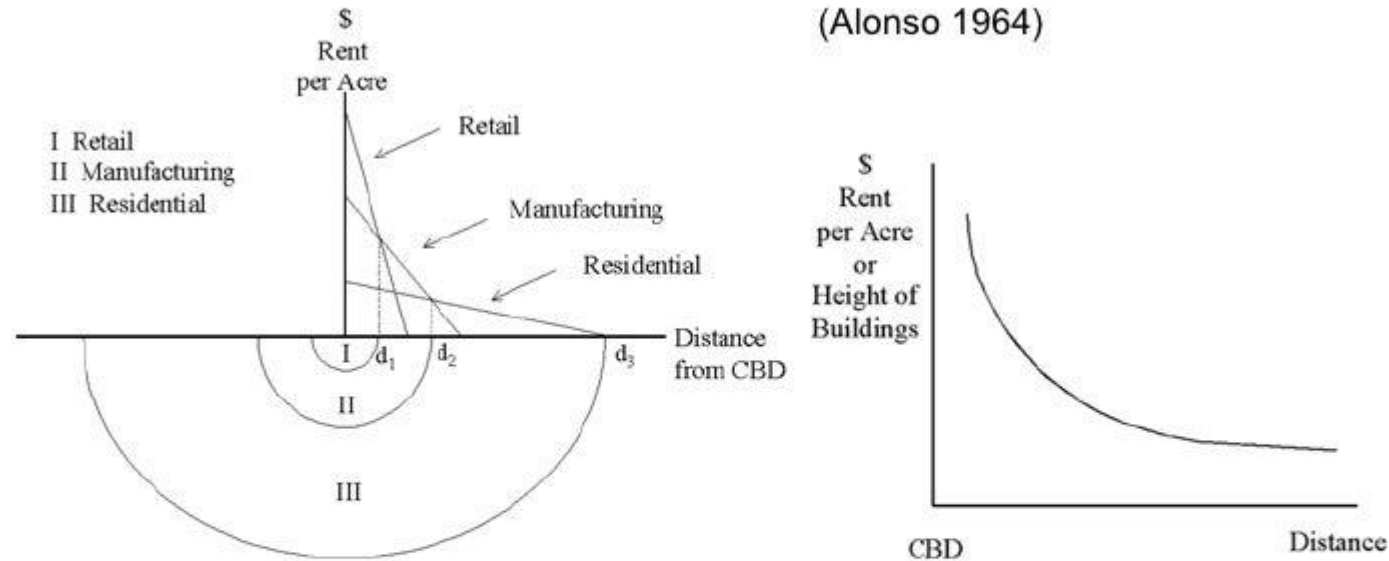
Von Thünen: urban land use



Alonso: bid-rent curve

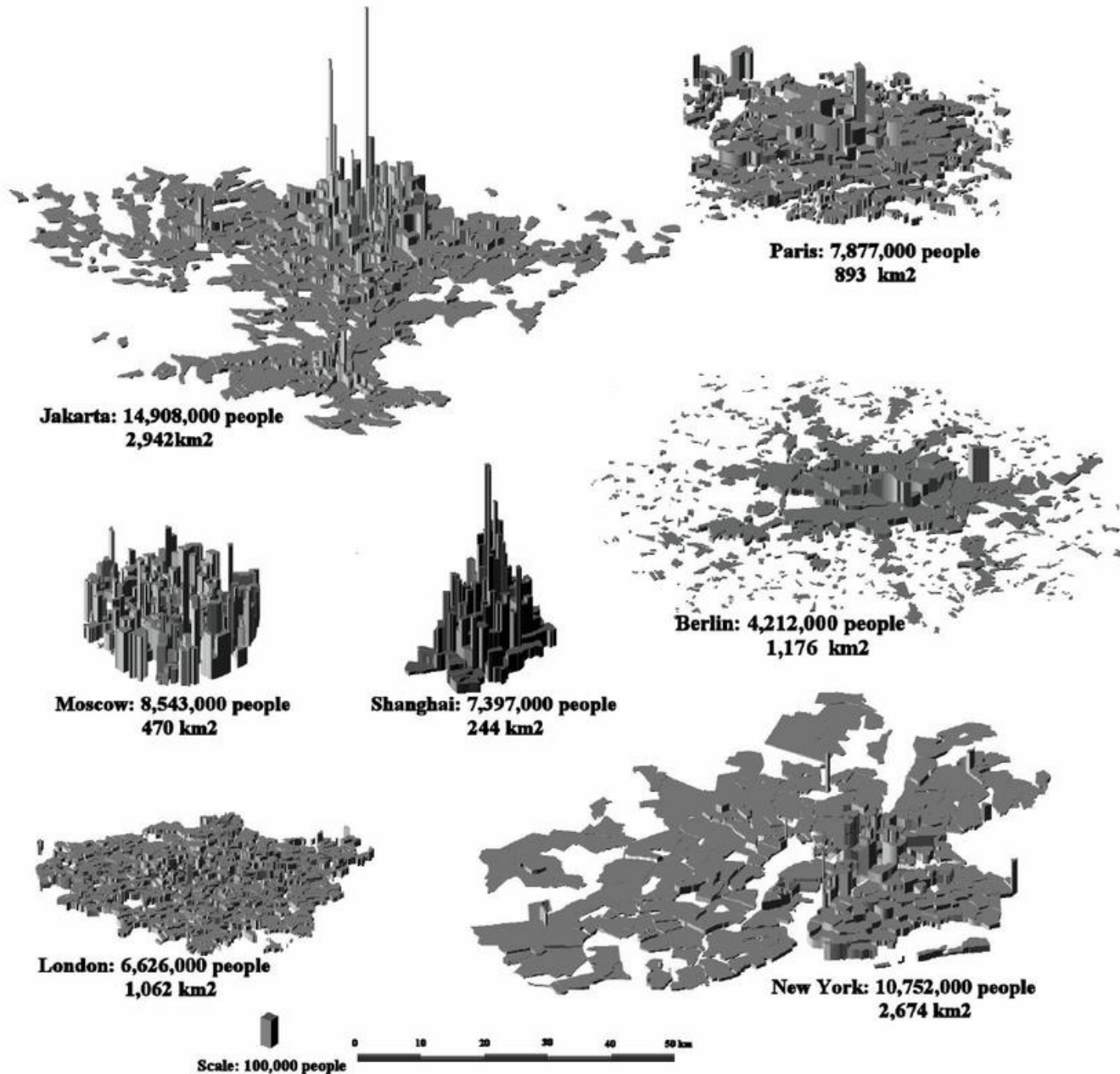
Bid Rent Theory

(Alonso 1964)

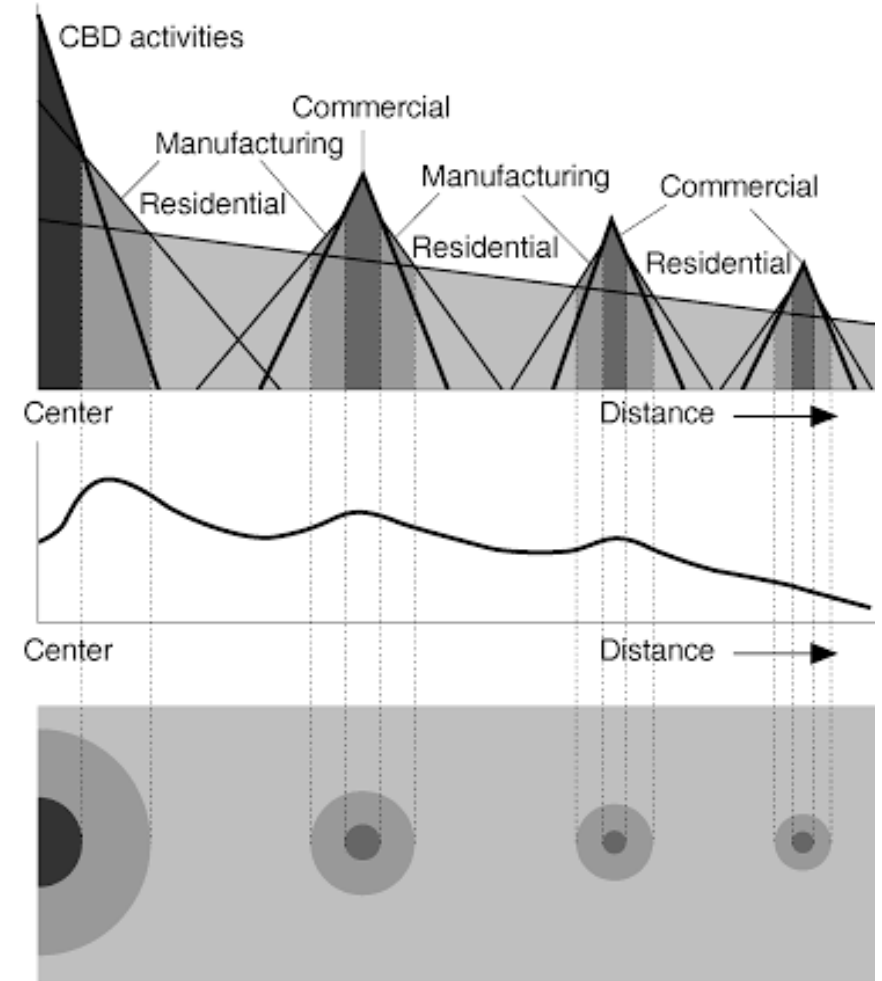


Bid rent/land rent theory shows how much different sectors of the economy are prepared to pay for land. Basic assumption is that accessibility is increased with centrality and therefore retailing is prepared to pay a high price for land in the CBD. As distance from the CBD increases availability of land increases and it is affordable for residential and even agricultural use.

Polycentric cities

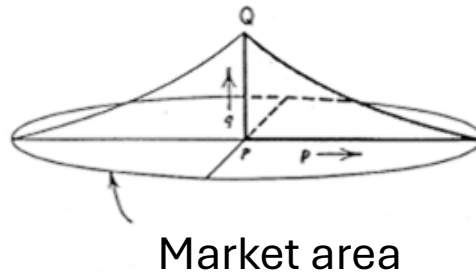


Polycentric

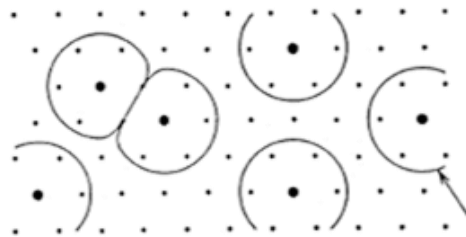


Christaller: how do centers emerge?

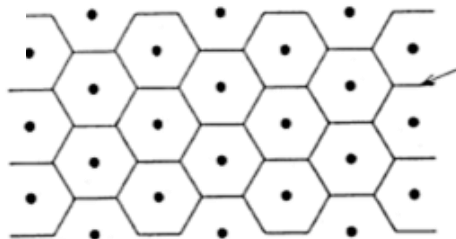
Phase 1



Phase 2



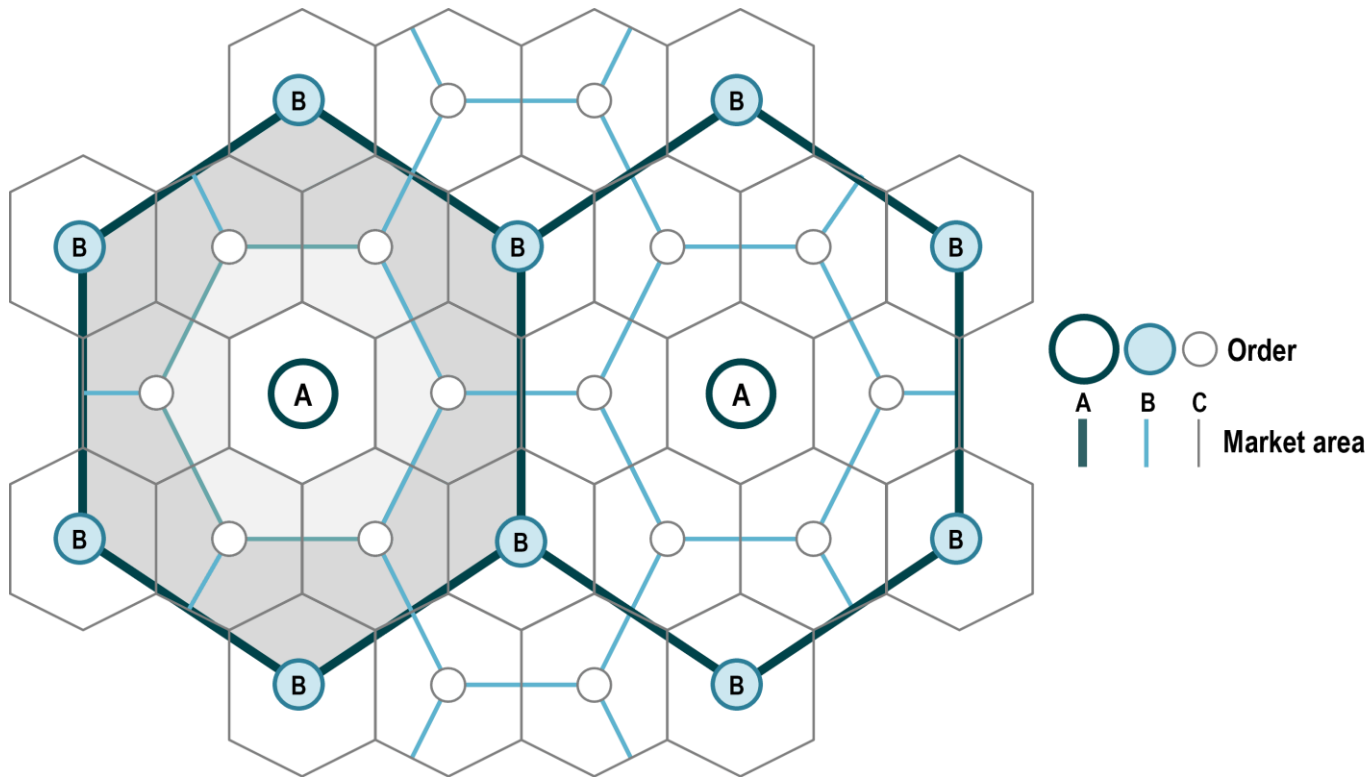
Phase 3



The Three Phases of Market Area Formation in a Single Industry:

1. **Single producer of goods** (natural market area)
2. **New firms enter** in the hope of earning extra profit, until supply covers the entire area. In overlapping zones, firms **compete** with one another.
3. A **long-term equilibrium** is established: all demand is satisfied, and natural market areas take on a **hexagonal shape**.

Christaller: central place theory



Derivation of the Urban Hierarchy:

- The **range** of services differ
- **Higher-order services** are only provided in **urban centers**
- **Higher-order centers** also provide **lower-order functions**

High-order center (A):

- College, university
- Specialized clinics

Mid-order center (B):

- Secondary education, vocational school
- Hospital, specialist doctors

Low-order centers (C):

- Primary education
- Pharmacy, general practitioner, dentist

Christaller: central place theory

Formation of the Urban Hierarchy

Lower-order services appear in **higher-level centers** due to **agglomeration advantages**.

The **range of lower-order services** is smaller and does not cover the entire area — leading to the establishment of **new locations**.

Three organizing principles:

(a) Market Principle:

A **minimal number** of lower-order centers are established so that the **entire surface is covered**.

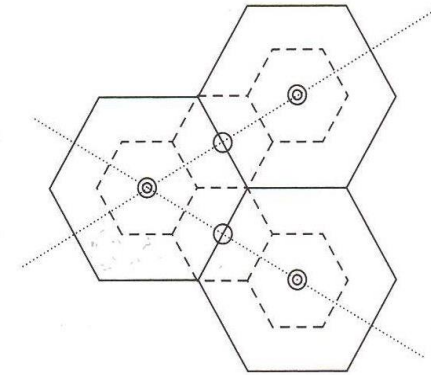
(b) Transportation Principle:

Lower-order centers are **optimally located** to **minimize transportation costs** from higher-order centers.

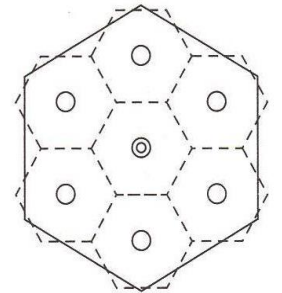
(c) Administrative Principle:

Each lower-order center is **clearly assigned** to a **specific higher-order center**.

(b) The transportation principle



(c) The administrative principle



3. Agglomeration economies

Definition and sources

Agglomeration Advantages:

- Firms become **more productive** thanks to agglomeration benefits.
- Growth is **faster in areas** where **positive externalities** are present.

Sources of agglomeration advantages (*Puga, 2010*):

- **Sharing:** use of labor and urban infrastructure
- **Matching:** meeting of supply and demand (e.g., employers and employees)
- **Learning:** knowledge spillovers and learning among firms

Static and dynamic agglomeration advantages:

- **Neo-Marshallian view:** local externalities improve firms' **static performance** (efficiency, cost)
- **Neo-Schumpeterian view:** local externalities enhance **innovative capacity**

Krugman: New Economic Geography

Explanation of Industry Concentration

- Increasing returns: stimulate agglomeration
- Transportation costs: encourage proximity to raw materials and intermediate goods
- Worker migration: increases labor supply and local market size



Paul Krugman (Princeton)
Nobel prize 2008

Krugman: centrum-periphery

Initially, equal distribution between the two regions: relocation of a firm has opposing effects

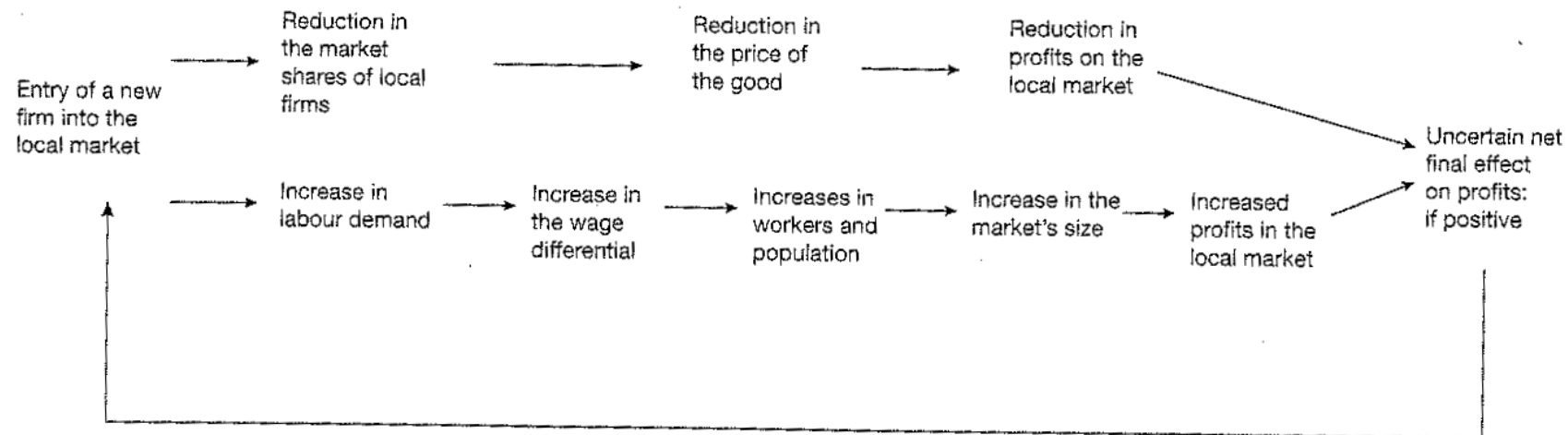
Competition effect:

- Depends on the **substitutability of products** (higher elasticity $\rightarrow$ greater profit loss)
- Also influenced by **transportation costs** (a decrease $\rightarrow$ lower prices $\rightarrow$ increased competition)

Demand effect:

- Driven by **increasing returns** (higher profit margins, stronger attraction for firms)
- Influenced by **income spent on products**

(a) The 'competition effect' and the 'demand effect'



High transportation costs:

- Stable equilibrium: firms do not relocate

Low transportation costs:

- Stable equilibrium: the industry concentrates in one region (where agglomeration advantages are greater)

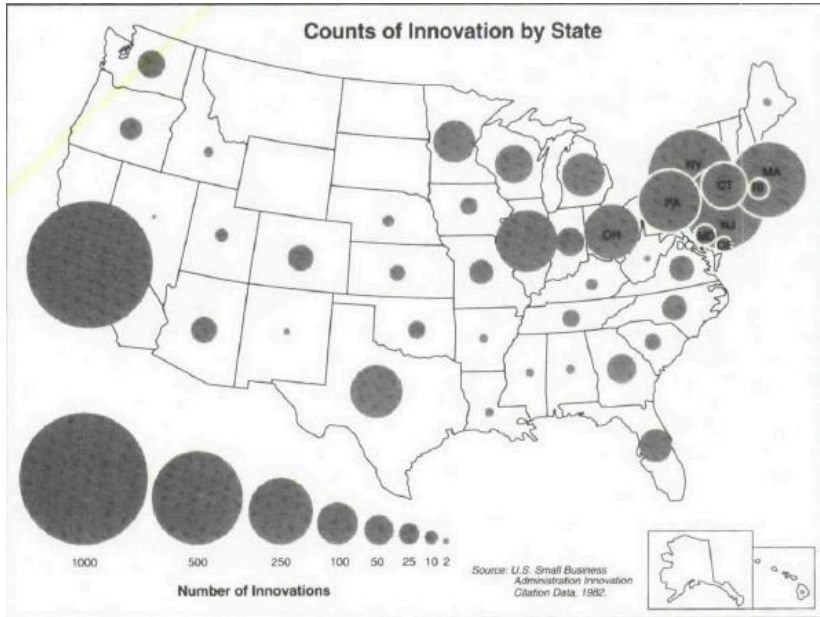
Moderately high transportation costs:

- Both equal distribution and core-periphery structure can be stable equilibria



4. Concentration of knowledge

Audretsch-Feldman 1996 – American Economic Review



- Concentration of technological innovation
- Spillovers are local
- Data: new products published in technological journals

- Gini index

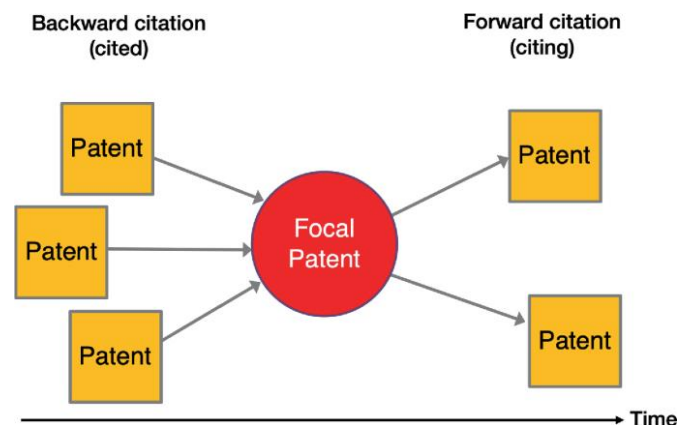
$$G = \frac{1}{2\bar{x}n^2} \sum_i \sum_j |x_i - x_j|$$

It compares the **average deviation** of an industry's share in a given region from its share in all other regions, **relative to the overall average**.

The larger the value, the more the industry is **concentrated in a few regions**.

- Comparison between production's and innovation's concentration
- Explanatory variables: natural resources, economies of scale, transportation costs, skilled workforce, R&D intensity, university R&D

Distance in knowledge spillovers



GEOGRAPHIC LOCALIZATION OF KNOWLEDGE SPILLOVERS AS EVIDENCED BY PATENT CITATIONS*

ADAM B. JAFFE
MANUEL TRAJTENBERG
REBECCA HENDERSON

We compare the geographic location of patent citations with that of the cited patents, as evidence of the extent to which knowledge spillovers are geographically localized. We find that citations to domestic patents are more likely to be domestic, and more likely to come from the same state and SMSA as the cited patents, compared with a “control frequency” reflecting the pre-existing concentration of related research activity. These effects are particularly significant at the local (SMSA) level. Localization fades over time, but only very slowly. There is no evidence that more “basic” inventions diffuse more rapidly than others.

The Quarterly Journal of Economics, Volume 108, Issue 3 (Aug., 1993), 577-598.



Journal of Urban Economics
Volume 42, Issue 3, November 1997, Pages 422-448



Regular Article

Local Geographic Spillovers between University Research and High Technology Innovations ☆

Luc Anselin ^a, Attila Varga ^{b c}, Zoltan Acs ^{d e}

Abstract

This paper re-examines the empirical evidence on the degree of spatial spillover between university research and high technology innovations. The familiar Griliches–Jaffe knowledge production function is estimated at both the state and the metropolitan statistical areas (MSA) level and extended with more precise measures of spatial spillover. Alternatives based on the gravity potential and covering indices are formulated for Jaffe's “geographical coincidence index” and found to provide strong evidence of local spillovers at the state level. At the MSA level, a distinction is made between research and development activities and university research in the MSA and in the surrounding counties. Evidence is found of local spatial externalities between university research and high technology innovative activity, both directly and indirectly via private research and development.

The gravity law

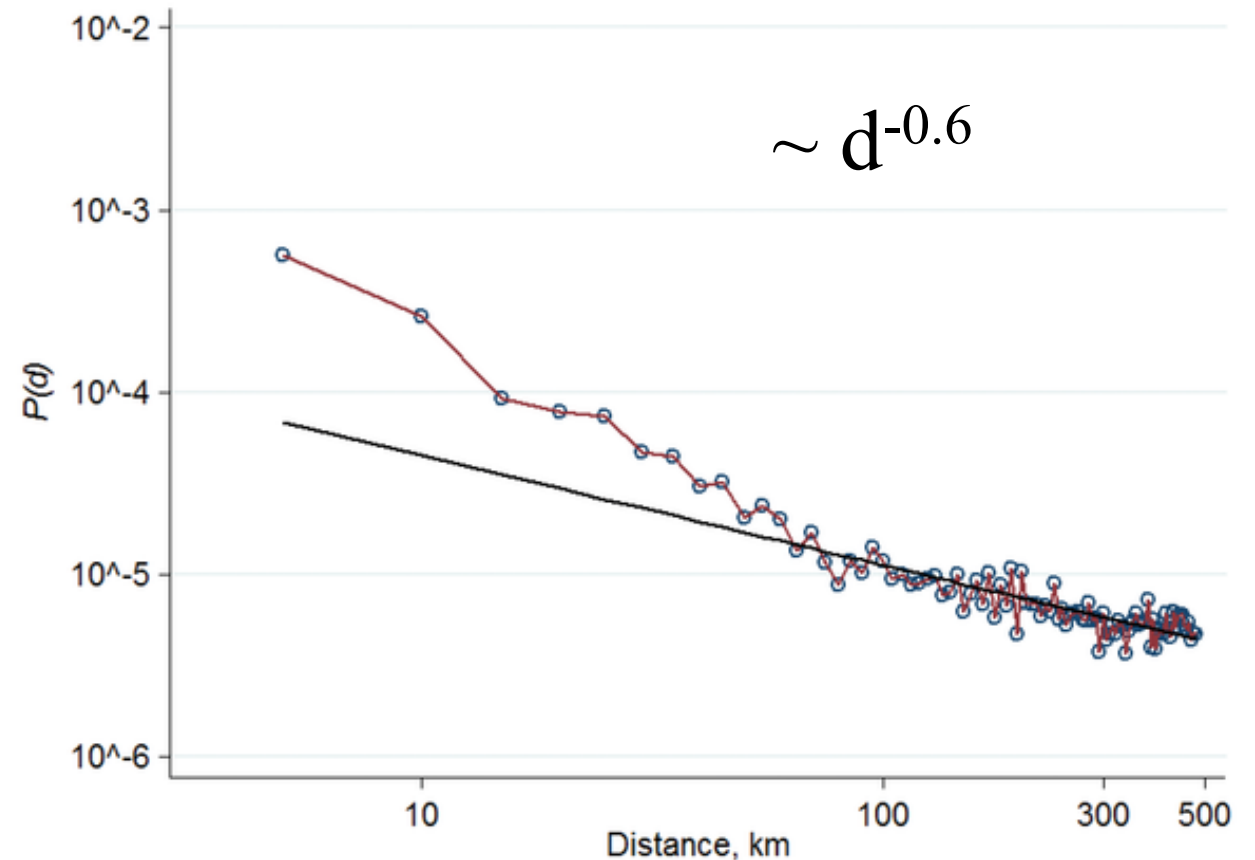
Zipf: $T_{ab} = K(P_a^\alpha P_b^\beta) / d_{ab}^\gamma$

Probability of relationships:

$N(d)$ – all possible connections

$L(d)$ – observed connections

$P(d) = L(d) / N(d)$



Urban scaling

$$Y(t) = Y_0 N(t)^\beta.$$

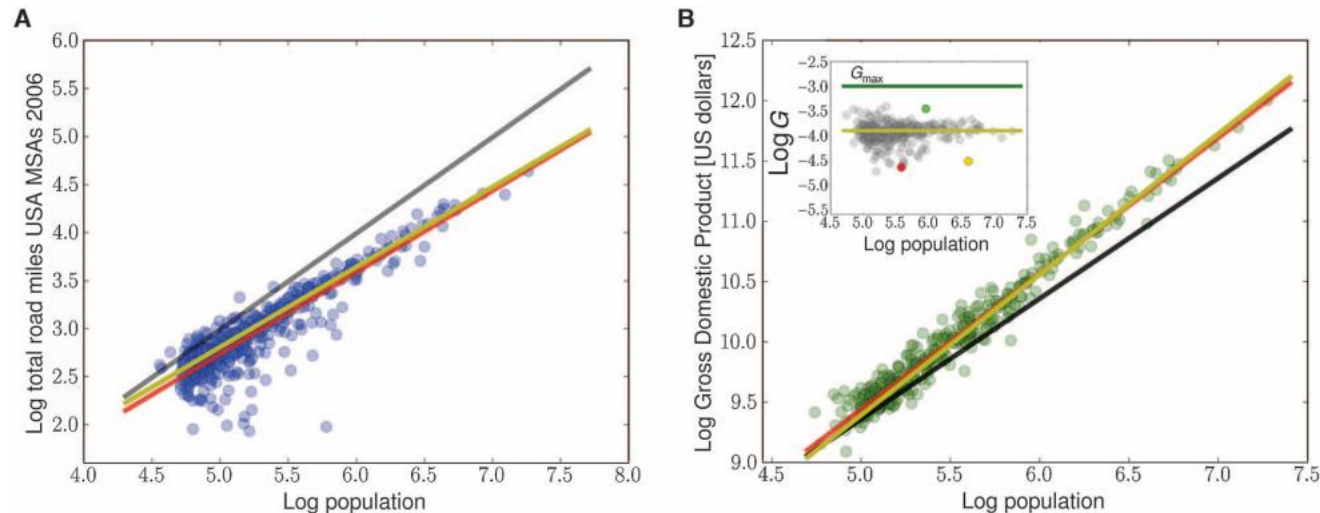


Fig. 1. Scaling of urban infrastructure and socioeconomic output. (A) Total lane miles (volume) of roads in U.S. metropolitan areas (MSAs) in 2006 (blue dots). Data for 415 urban areas were obtained from the Office of Highway Policy Information from the Federal Highway Administration (14). Lines show the best fit to a scaling relation $Y(N) = Y_0 N^\beta$ (red), with $\beta = 0.849 \pm 0.038$ [95% confidence interval (CI), $R^2 = 0.65$]; the theoretical prediction, $\beta = 5/6$ (yellow); and linear scaling $\beta = 1$ (black). **(B)** Gross metropolitan product of MSAs in 2006 (green dots). Data obtained for 363 MSAs from U.S. Bureau of Economic Analysis (14). Lines describe best fit (red) to data, $\beta = 1.126 \pm 0.023$ (95% CI, $R^2 = 0.96$); the theoretical prediction, $\beta = 7/6$ (yellow); and proportional scaling, $\beta = 1$ (black). The two best-fit parameters in each scaling

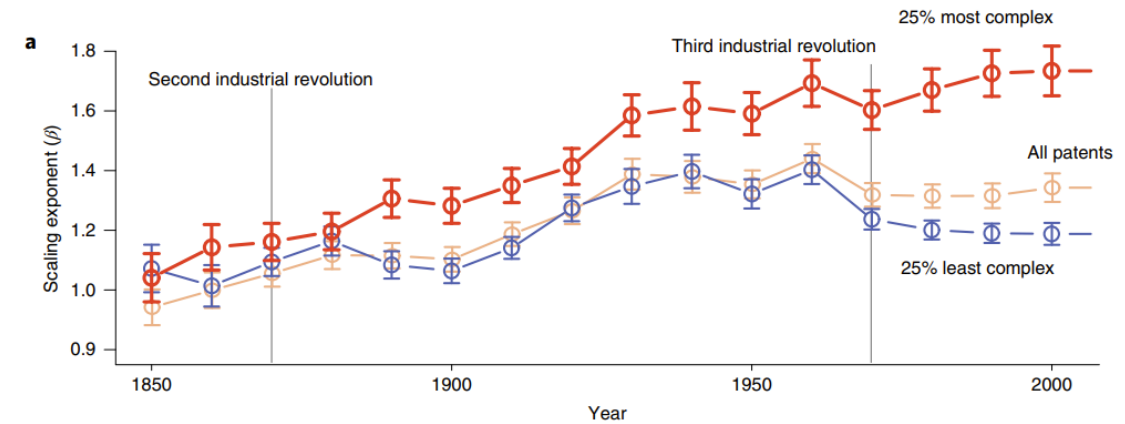
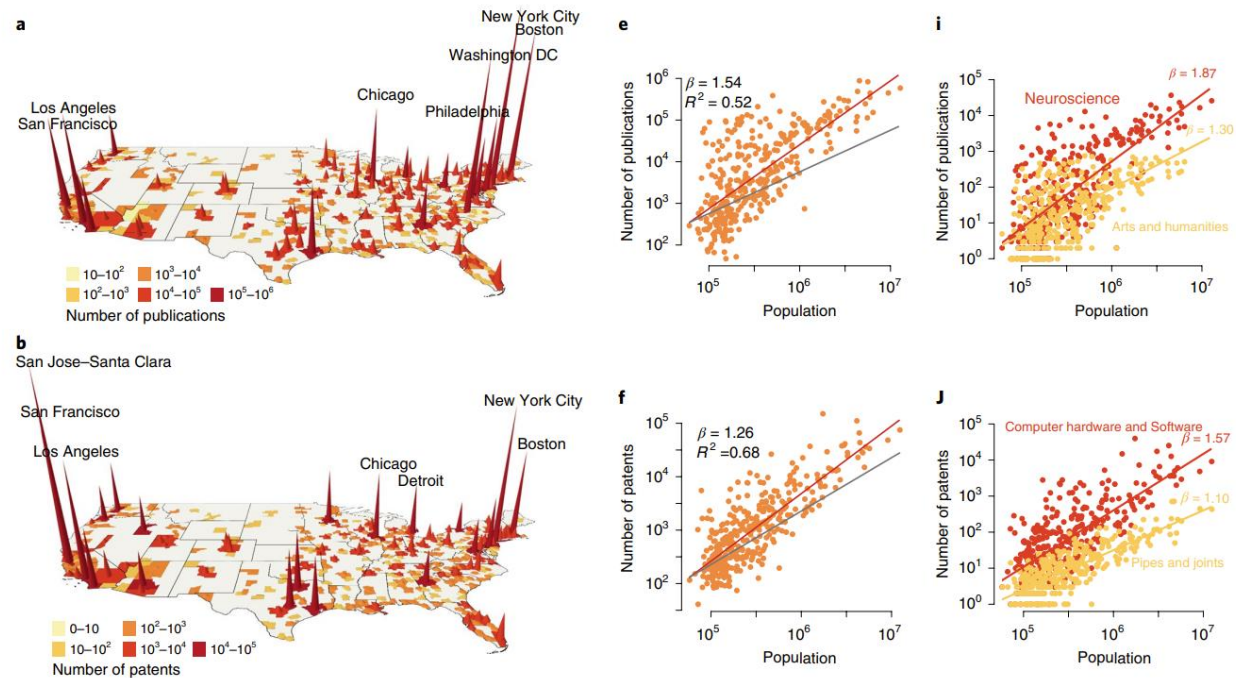
relation were estimated by means of ordinary least-squares minimization to the linear relation between logarithmically transformed variables (14). The inset shows the estimate of G for 313 U.S. MSAs and the conservation law $\frac{d \ln G}{d \ln N} = 0$ ($R^2 = 0.003$). G is measured as the product of gross domestic product and road volume, both per capita. As predicted by the theory, observed values of G for different cities cluster around its most likely value (mode, yellow line), which gives an estimate of the optimum G^* , and are bounded by the maximum $G_{\max} \approx 8G^*$ (green line); see also Fig. 2B. Several metropolitan areas, such as Bridgeport, Connecticut (green circle); Riverside, California (yellow circle); or Brownsville, Texas (red circle), are outliers, suggesting that they are suboptimal in terms of their transportation efficiency or amount of social mixing.

Table 1. Scaling exponents for urban indicators vs. city size

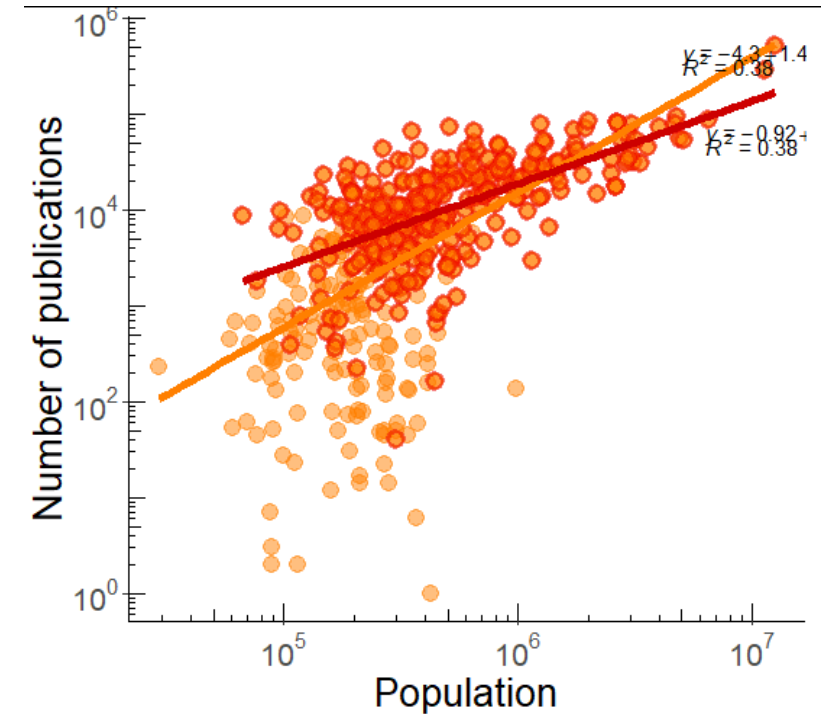
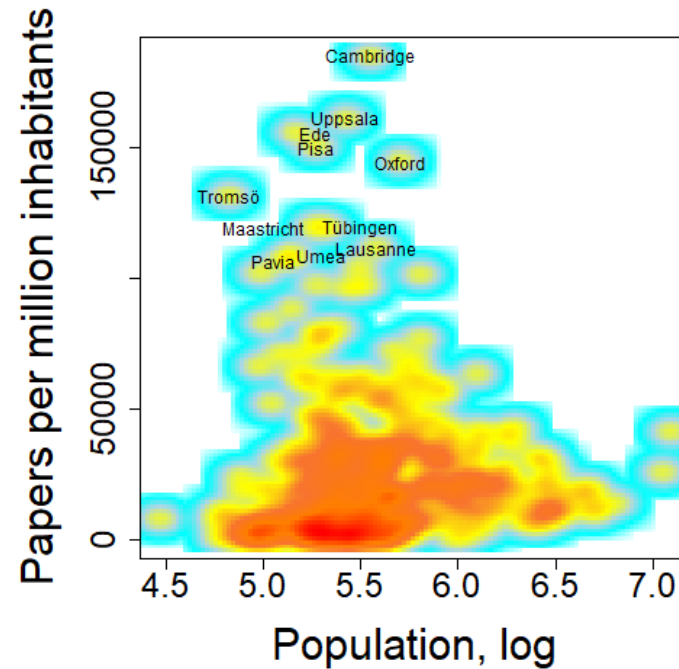
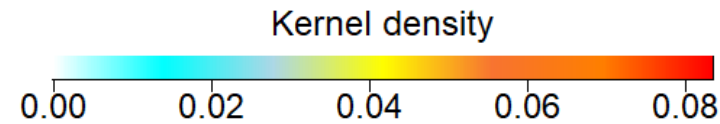
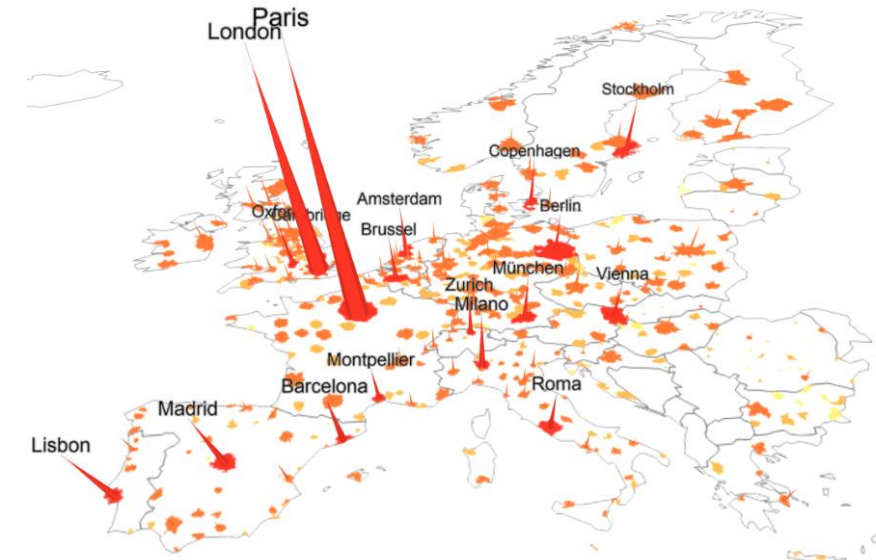
Y	β	95% CI	Adj- R^2	Observations
New patents	1.27	[1.25, 1.29]	0.72	331
Inventors	1.25	[1.22, 1.27]	0.76	331
Private R&D employment	1.34	[1.29, 1.39]	0.92	266
"Supercreative" employment	1.15	[1.11, 1.18]	0.89	287
R&D establishments	1.19	[1.14, 1.22]	0.77	287
R&D employment	1.26	[1.18, 1.43]	0.93	295
Total wages	1.12	[1.09, 1.13]	0.96	361
Total bank deposits	1.08	[1.03, 1.11]	0.91	267
GDP	1.15	[1.06, 1.23]	0.96	295
GDP	1.26	[1.09, 1.46]	0.64	196
GDP	1.13	[1.03, 1.23]	0.94	37
Total electrical consumption	1.07	[1.03, 1.11]	0.88	392
New AIDS cases	1.23	[1.18, 1.29]	0.76	93
Serious crimes	1.16	[1.11, 1.18]	0.89	287
Total housing	1.00	[0.99, 1.01]	0.99	316
Total employment	1.01	[0.99, 1.02]	0.98	331
Household electrical consumption	1.00	[0.94, 1.06]	0.88	377
Household electrical consumption	1.05	[0.89, 1.22]	0.91	295
Household water consumption	1.01	[0.89, 1.11]	0.96	295
Gasoline stations	0.77	[0.74, 0.81]	0.93	318
Gasoline sales	0.79	[0.73, 0.80]	0.94	318
Length of electrical cables	0.87	[0.82, 0.92]	0.75	380
Road surface	0.83	[0.74, 0.92]	0.87	29

Complex economic activities concentrate in large cities

Pierre-Alexandre Balland ^{1,2*}, Cristian Jara-Figueroa ², Sergio G. Petralia ³, Mathieu P. A. Steijn ^{1,4}, David L. Rigby ⁵ and César A. Hidalgo ^{2,6,7,8}



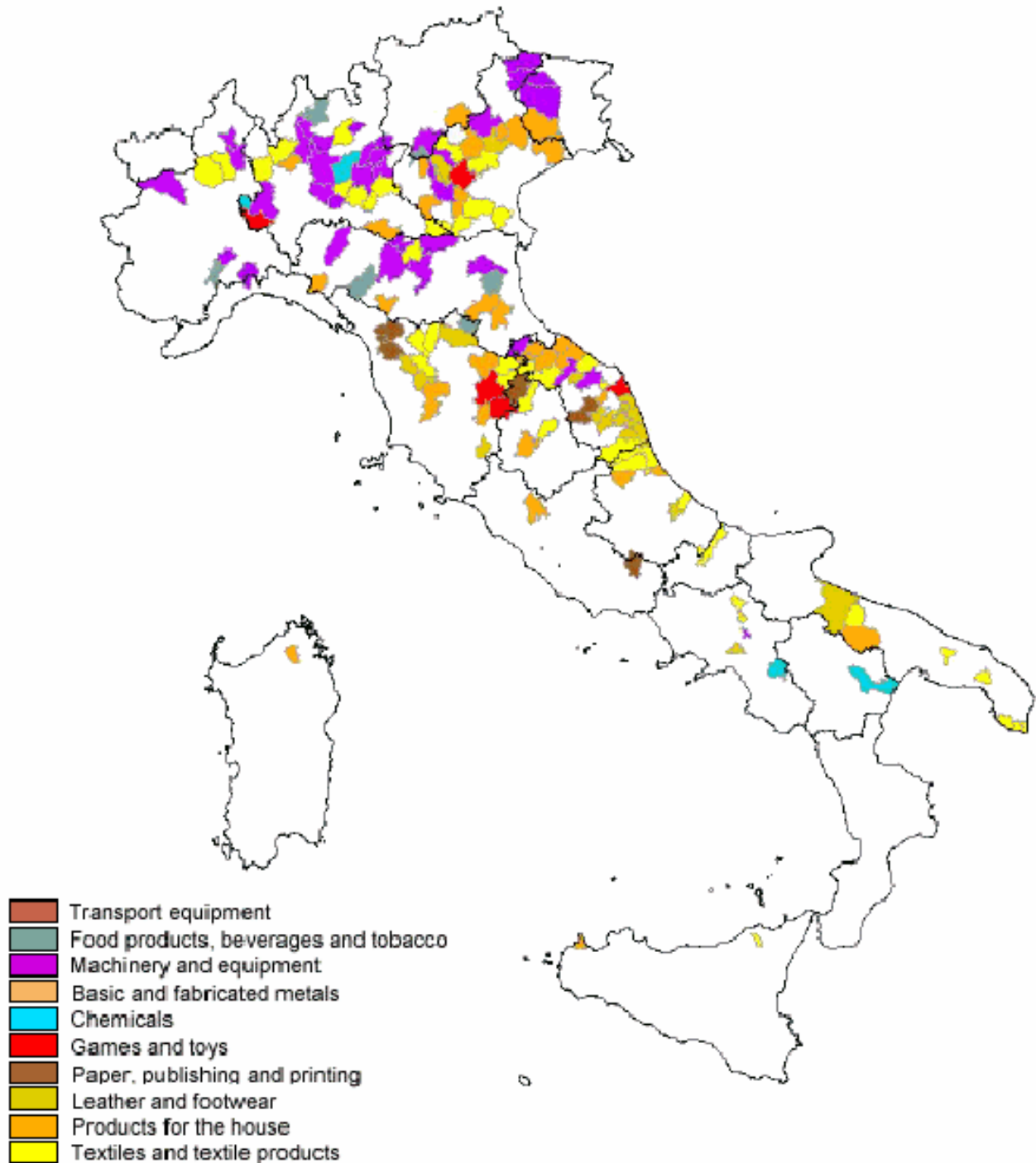
Sacling of European university knowledge is different



5. Specialization versus diversity

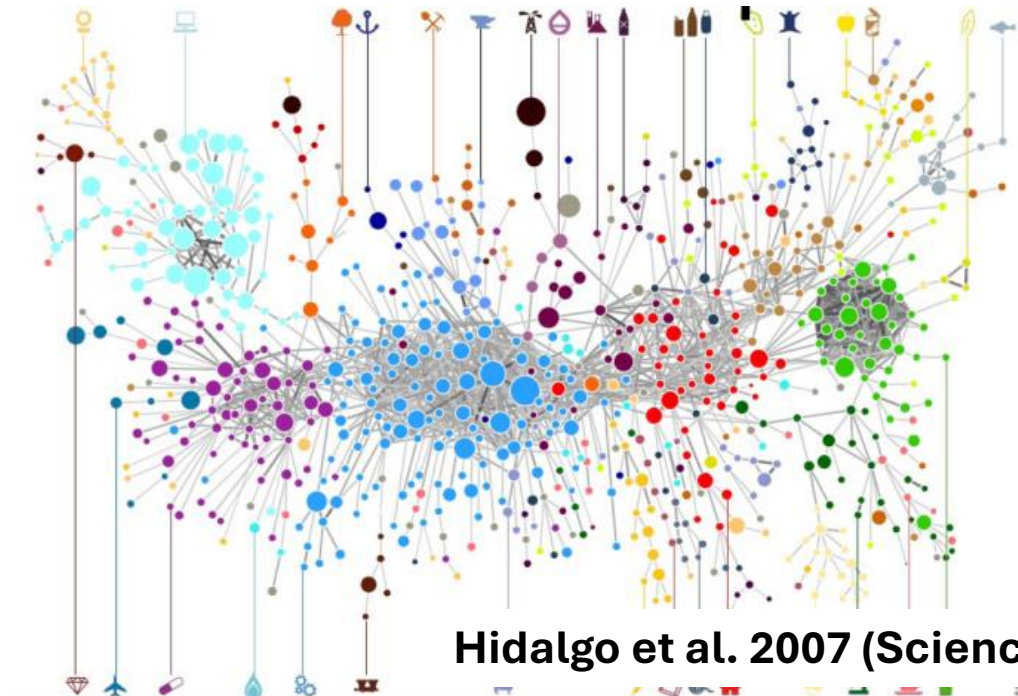
Italian industrial districts

- SMEs (Small and Medium-sized Enterprises)
- Local traditions and strong cohesion
- Craft industry heritage
- Small town and its surroundings
- Business development organizations
- Supportive local governments



Economic progress and urban success

- Social interaction:
population density
- Learning in the city:
industry structure
- Related knowledge is
easier to learn but
contains less novelty



Hidalgo et al. 2007 (Science)

Specialization: export base

Location Quotient

- Indicates the **degree of specialization** of a region in a given industry.
- Compares the **local share of employment** in the industry to the **national average**.
- Assumption: the **specialized product of the region is not only consumed locally**, thus the region participates in **interregional trade**.

$$LQ = \frac{\frac{E_{ir}}{E_r}}{\frac{E_{in}}{E_n}}$$



Diversity

Cities are arenas of interaction

Mixing in the built environment

